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A STUDY OF THE DYNAMICS OF TRIPLE SYSTEMS BY MEANS OF STATISTICAL SAMPLING

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Trajectories in the plane problem of three bodies with equal masses have been calculated on an electronic computer. Altogether 100 different initial conditions are examined. The initial configurations were assigned by means of random numbers. The initial velocities of the component bodies were assumed to have the same small values in all cases. For 62 of the initial conditions the computations were carried to completion, and their accuracy is found satisfactory. In all these the motion was concluded by disintegration of the system. In every instance disintegration occurred after a close triple approach of the components. The mean disintegration time for a triple system (having negative energy and a small angular momentum), as computed from the results for 59 systems, is 62 ± 10 time units, where the unit is the mean time required for a component to cross the system. This corresponds to a mean disintegration time of $(1.6 \pm 0.3) \cdot 10^6$ yr for triple systems having masses of one solar mass and a mean distance of 0.03 pc between components. A triple galaxy with components of mass $2 \cdot 10^{43}$ g and a mean distance of 30 kpc between components would disintegrate, on the average, in 16 billion years. A diagram illustrates the motions of the components in one of the systems investigated, from the start of the motion until disintegration.

During the past few years several authors [1-8] have successfully used high-speed electronic computers for numerically integrating systems of equations of motion in the n-body problem. The possibilities for adopting large values of n in these integrations have been limited by the capacity of electronic computers to solve systems of high-order differential equations and by the rapid lengthening of the necessary operating time on a computer as n increases. In the investigations mentioned the authors have adopted as large a value for n as possible. In particular, computations have been performed for the value n = 100 [4, 6, 8], which enables the results to be applied to such actual groups of bodies as open star clusters and moderately populous clusters of galaxies.

However, the lengthy computer operating time that is demanded if n is large sharply restricts the number of cases having different initial conditions, and as a consequence we still have an inadequate understanding of the dynamics of open star clusters and clusters of galaxies. As a rule only one choice of initial conditions has been treated by previous investigators.

Considerably wider possibilities are afforded in this respect if n does not have a large value, especially if n is equal to three. Triple systems are recognized to be widely distributed among stars and galaxies. Thus in addition to its general celestial-mechanics interest, a study of the dynamics of triple systems also offers much interest for stellar astronomy and cosmogony.

Let us formulate, then, the problem of investigating the dynamics of triple systems by means of an electronic computer. We shall attempt to cover a representative sample from the infinite set of different initial conditions for a triple system.

In the general three-body problem two numbers must be prescribed at initial time in order to specify the configuration of the system. Six numbers must be assigned to establish the initial velocities of the bodies, because the nine velocity

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1968SvA....11.1006A

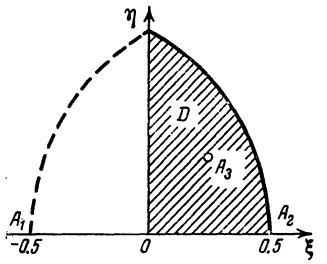


Fig. 1

components will naturally be related by three conditions for immobility of the center of inertia of the system. Thus in the general problem of these bodies eight numbers in all must be prescribed to specify the initial state of the system.

In order to reduce the necessary operating time we shall in this investigation consider the plane three-body problem in which the masses of all three bodies are assumed equal.

Let the bodies A_1, A_2, and A_3 be located at the points $(-\frac{1}{2}, 0)$, $(+\frac{1}{2}, 0)$, and (ξ, η) respectively. From point A_1 we draw a circular arc of unit radius. By allowing the point A_3 to move over the shaded region D in Fig. 1, we can obtain all possible configurations for the triple system.

We shall express distances in the units $\rho = 1 = 3 \cdot 10^{16} l$ cm, and for masses $m = 1 = 2 \cdot 10^{33}$ k g. The masses of the components of the system will be considered equal to this unit mass. The gravitation constant is $6.67 \cdot 10^{-8}$. These units will then define a unit of time

$$t = 1 = \frac{\rho^{3/2}}{m^{1/2}} = \sqrt{1.35 \cdot 10^8} \cdot \sqrt{\frac{l^3}{k}} \text{ sec}, \qquad (1)$$

and a unit of velocity

$$v = 1 = \frac{3}{\sqrt{1.35}} \cdot 10^8 \cdot \sqrt{\frac{k}{l}} \text{ cm/sec}. \qquad (2)$$

Trial computations performed for zero initial velocities of the components have shown that in this situation the very first approach of the components would normally be so close that the computation process would be severely complicated. For this reason the velocity components for the bodies A_1, A_2, and A_3 in each instance were taken at initial time to be

$$\begin{aligned} \dot{x}_1 &= +0.00005, & \dot{y}_1 &= 0, \\ \dot{x}_2 &= +0.00005, & \dot{y}_2 &= 0, \\ \dot{x}_3 &= -0.0001, & \dot{y}_3 &= 0. \end{aligned} \qquad (3)$$

Under this condition the kinetic energy of the system at initial time equal to $7.5 \cdot 10^{-9}$ would be at least $26\frac{2}{3}$ times smaller than the absolute value of the potential energy, for which, since $A_1A_2 = 1$, $A_1A_3 \leq 1$, $A_2A_3 \leq 1$, we have the inequality

$$|U| \geqslant 3 \cdot 6.67 \cdot 10^{-8}.$$

This circumstance implies that the angular momentum of the system is small.

We introduce now the concepts of the mean size a of the triple system, the rms velocity $\bar{v}$ in the system, and the mean time τ for crossing the system. For this purpose we assume that in some triple system B (whose components have unit mass) which has the same total energy E as the system under consideration, the conditions of the virial theorem are satisfied at a certain time t', and the configuration is an equilateral triangle. We shall call the length a of a side of this triangle the mean size of the system under consideration. The quantity $\bar{v}$ will be taken as the rms velocity of the components in the system B at time t' and τ will be defined as $a/\bar{v}$. If we then denote the kinetic energy of the system in question at initial time by L_0, we have

$$-\frac{3G}{2a} = -\frac{3\bar{v}^2}{2} = E = -G\left(1 + \frac{1}{\sqrt{\left(\xi + \frac{1}{2}\right)^2 + \eta^2}} + \frac{1}{\sqrt{\left(\xi - \frac{1}{2}\right)^2 + \eta^2}}\right) + L_0 \qquad (4)$$

and consequently

$$a = \frac{3}{2}\left[1 + \frac{1}{\sqrt{\left(\xi + \frac{1}{2}\right)^2 + \eta^2}} + \frac{1}{\sqrt{\left(\xi - \frac{1}{2}\right)^2 + \eta^2}} - \frac{L_0}{G}\right]^{-1}, \qquad (5)$$

$$\bar{v} = \left[\frac{2}{3}G\left(1 + \frac{1}{\sqrt{\left(\xi + \frac{1}{2}\right)^2 + \eta^2}} + \frac{1}{\sqrt{\left(\xi - \frac{1}{2}\right)^2 + \eta^2}} - \frac{L_0}{G}\right)\right]^{\frac{1}{2}}, \qquad (6)$$

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1968SvA....11.1006A

$$\tau = G^{-\frac{1}{2}}\left[\frac{2}{3}\left(1+\frac{1}{\sqrt{\left(\xi+\frac{1}{2}\right)^2+\eta^2}}+\frac{1}{\sqrt{\left(\xi-\frac{1}{2}\right)^2+\eta^2}}-\frac{L_0}{G}\right)\right]^{-\frac{3}{2}} \quad (7)$$

in terms of the units adopted.

Equations (5)-(7) show that for different initial conditions a, $\bar{v}$, and τ contain a differing number of our adopted units. By introducing the quantities a, $\bar{v}$, and τ we ensure a unified approach to systems having different initial conditions.

A program for generating pairs of pseudo-random numbers on the M-20 and BÉSM-3M computers has been adopted from Lyashenko's book [9]. To obtain ξ and η, the pseudo-random numbers generated were multiplied by $1/2$ and $3/2$ respectively and the condition

$$\left(\xi+\frac{1}{2}\right)^2+\eta^2 \ll 1 \quad (8)$$

was tested to determine whether the point fell into the region D. If the condition (8) was not satisfied, then the trajectories of the bodies were not computed for the given pair ξ, η.

Computations of the trajectories of the bodies in the system were performed for 100 successive pairs of initial values ξ, η satisfying the condition (8). The numerical solution of the system of equations of motion

$$\left.\begin{aligned} \ddot{x}_i &= -G\sum_{k\neq i}\frac{x_i-x_k}{r_{i,k}^3}, \quad \ddot{y}_i = -G\sum_{k\neq i}\frac{y_i-y_k}{r_{i,k}^3} \\ i &= 1,2,3, \quad r_{i,k} = [(x_i-x_k)^2+(y_i-y_k)^2]^{1/2} \end{aligned}\right\} \quad (9)$$

on the computer utilized the Runge-Kutta procedure with an automatic step. The automatic step was selected in accordance with the requirement

$$0.1 < \left|\frac{\delta x}{\varepsilon_p}\right| < 1,$$

where ε_p is the prescribed accuracy of the computation, and

$$\delta x = \frac{2}{3}h(f_{i,0}-f_{i,1}+f_{i,2}-f_{i,3}),$$

$$f_{i,0} = f(x_i), \quad f_{i,1} = f\left(x_i+\frac{h}{2}f_{i,0}\right),$$

$$f_{i,2} = f\left(x_i+\frac{h}{2}f_{i,1}\right), \quad f_{i,3} = f(x_i+hf_{i,2}),$$

with f the function in the right-hand member of the equations of motion.

In order that the ratio of the change in the energy of the system to its value at the initial time would not exceed 0.05, the value $\varepsilon_p = 0.25 \cdot 10^{-6}$ was adopted for the greater part of the computations, while $\varepsilon_p = 0.5 \cdot 10^{-7}$ was used for the remaining systems.

As a check on the computations we have used the constancy of the total energy of the system, the constancy of the total angular momentum, and the condition that the center of inertia of the system be stationary. The check was performed at each step. Moreover, each check was performed twice.

The question of whether a triple system having negative energy and not subject to external influences will disintegrate has hitherto remained unresolved. Tevzadze [10] gives an example of a numerical solution to the equations of motion of a system which indicates the possibility of disintegration. But in this example the initial conditions are specially stipulated so as to favor disintegration. Thus even though the possibility of disintegration of a triple system has been demonstrated by Tevzadze, it has remained unclear whether disintegration would occur in the general case of the motion.

The calculations performed in the present investigation have in fact shown that, at least for small values of the angular momentum of the system and in the case of plane motion, disintegration will occur for any initial states of the system.

Of the 100 computations performed, 64 have been extended to the time when disintegration of the system takes place. In order to determine whether disintegration has occurred, we have applied Tevzadze's criterion [10], which states that at a certain time the conditions

$$R \geqslant \frac{2G\left(\sum_{i=1}^{3} m_i\right)^2}{3|E|}, \qquad v_j^2 \geqslant \frac{2G\left(\sum_{i=1}^{3} m_i - m_j\right)}{\sqrt{x_j^2+y_j^2}} \quad (10)$$

must be satisfied, where j is the number of the component receding from the other two, R is the distance of this component from the center of inertia of the other two components, and v_j is the velocity of the j-th component. As Tevzadze has shown, if the conditions (10) are satisfied then $R(t) \to \infty$ as $t \to \infty$. In the event that Tevzadze's criterion is satisfied

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1968SvA....11.1006A

we have called the disintegration "exact." However, cases can occur where the criterion (10) is not satisfied, but nevertheless where one of the components recedes arbitrarily far from the other two. Correspondingly, both the recession and the subsequent return of this component may take an unlimited amount of time. Thus if Tevzadze's criterion is not satisfied but if the distance of one of the components from the center of inertia reaches 10 units, we have considered that a "nominal" disintegration has taken place. This terminology may be justified by the circumstance that if a component of a triple star recedes to a comparatively large distance under real conditions it will, as a rule, be detached from the system by regular or irregular forces of the general star field. In both the case of exact disintegration and the case of nominal disintegration the time of the disintegration is regarded as the epoch of closest approach of the bodies, following which the conditions (10) are observed to be satisfied, or one of the components has receded to a distance of 10 units.

In 21 cases the computations did lead to a disintegration but they were not considered satisfactory because after disintegration the energy of the system differed by more than 5% from the original energy. Evidently in these cases such close approaches occur during the motion of the components that the computation procedure we have employed does not ensure a sufficiently small value for the quantity $|\Delta E/E|$. These cases require changes in the computation method.

In the 17 remaining cases the computations could not be extended to completion since an approach with $r_{i,k} < 3 \cdot 10^{-7}$ occurred in the course of the motion, which essentially resulted in a division by zero in Eqs. (9), so that the computer stopped. These cases also call for changes in the computation procedure.

In all the examples computed disintegration of the system occurred after a triple, but not after a double approach of the components. This result appears important to us. Evidently disintegration is possible only after a triple approach with

$$p < \lambda a, \tag{11}$$

where p is the minimum perimeter of the configuration triangle during the approach, and λ is a suitable constant. On the basis of our results λ may be considered close to 0.3.

Table 1 presents the results of the computations. Column 1 contains the serial numbers of the initial conditions as specified by a sequence of random numbers. The corresponding values of ξ and η are entered in columns 2 and 3. If the computation was not carried to completion the reason is indicated in column 4. Column 5 gives the disintegration time T, that is, the elapsed time from the initial epoch to the epoch of disintegration. It is expressed in units of the period τ, as defined by Eq. (7). In column 6 is entered the value of p for the triple approach after which disintegration of the system has occurred, in units of the mean size a of the system. The computer recognized triple approaches with $p \leq 0.60$ units of distance ρ. In three cases where nominal disintegration of the system occurred, the last triple approach was not recognized (that is, p was greater than 0.60), so that the epochs of decay and the value of p were not determined for these systems. If a given system suffered no approach closer than the one that resulted in disintegration, then a dash appears in column 7; otherwise the value of p is indicated for the closest approach. Column 8 gives E, and column 9 $\Delta E/E$, where ΔE is the change in energy through accumulation of errors in the calculation. Column 10 contains $\Delta'E/E$, where $\Delta'E$ is the excess in energy over that required by Tevzadze's criterion for the component leaving the system. The angular momentum M of the system is given in column 11, and the quantity $\Delta M/M$ in column 12, where ΔM is the change in the angular momentum due to accumulation of errors in the calculation.

As the table shows, in 56 of 62 cases $\Delta E/E < 0.03$, and for 43 triple systems $\Delta E/E < 0.01$. The quantity $\Delta M/M$ remains very small, not exceeding 0.0001. Column 6 indicates that in 49 of the 62 cases disintegration occurred as a result of a very close triple approach. This circumstance shows that the closer the triple approach, the more probable disintegration will be. A comparison of columns 9 and 10 demonstrates that as a rule (in 33 of 48 cases) $\Delta'E > \Delta E$. Hence there is no reason to believe that the disintegration of the systems results from an accumulation of errors of computation. The disintegration of triple systems is a physical phenomenon.

As an illustration, Figs. 2a-2h display the trajectories for the motion of the components of a triple system having the initial configuration $\xi = 0.269722548$, $\eta = 0.476622422$. Each successive graph represents a continuation of the previous one, each curve in a graph (solid, dashed, and dot-dash) indicates the trajectory for the motion of one component of the system, the arrows designate the direction of motion, and the numbers indicate moments of time, expressed in units of τ. We see that after

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1968SvA....11.1006A

TABLE 1

N	ξ	η		T	p	p_{min}	$E\cdot10^{-7}$	$\frac{\Delta E}{E}$	$\frac{\Delta' E}{E}$	$M\cdot10^{-4}$	$\frac{\Delta M}{M}\cdot10^{-4}$
1	2	3	4	5	6	7	8	9	10	11	12
1	0.152107965	0.516951804	Disintegration	24.7	0.0707	—	—2.254	0.026	—0.842	0.5170	0.0454
2	0.022976316	0.521562782	Disintegration	93.7	0.0428	—	—2.301	0	0.479	0.5216	0.1304
3	0.269722548	0.476622422	Disintegration	26.0	0.1145	—	—2.041	0	0.283	0.4766	0.0125
4	0.302172674	0.431941366	Disintegration	315.9	0.0509	—	—1.836	0.020	1.479	0.4319	0.1364
5	0.005310069	0.666154059	Disintegration	214.3	0.2007	0.1612	—2.843	0.040	0	0.6662	0.1048
6	0.184999239	0.600560507	Disintegration	1.9	0.1233	—	—2.570	0.001	0.438	0.6006	0.0109
7	0.005159629	0.385816848	Nominal disinteg.	161.6	0.1722	0.0632	—1.804	0.034	—	0.3858	0.3227
8	0.157232181	0.393068735	Disintegration	2.1	0.0848	—	—1.817	0.001	1.041	0.3931	0.0143
9	0.032952633	0.367271030	Disintegration	104.7	0.0447	—	—1.813	0.013	0.980	0.3673	0.0485
10	0.248615885	0.122069321	Disintegration	22.8	0.0269	—	—0.887	0.010	0.898	0.1221	0.2875
11	0.425790152	0.158446944	$\frac{\Delta E}{E}>0.05$	—	—	—	—	—	—	—	—
12	0.426903973	0.236476070	Disintegration	55.2	0.0498	—	—0.836	0.012	0	0.2365	0.4465
13	0.282590939	0.617363867	Close approach	—	—	—	—	—	—	—	—
14	0.189034578	0.099451113	Close approach	—	—	—	—	—	—	—	—
15	0.008491508	0.593936046	Disintegration	175.7	0.1737	0.0766	—2.564	0.005	0	0.5939	0.0477
16	0.078425390	0.349994402	Disintegration	129.6	0.1053	—	—1.747	0.003	0.4776	0.3490	0.0169
17	0.1152/9340	0.749806025	Disintegration	1.9	0.2422	—	—3.185	0	0	0.7498	0.0060
18	0.012014848	0.514189941	Close approach	—	—	—	—	—	—	—	—
19	0.041515318	0.353444473	Close approach	—	—	—	—	—	—	—	—
20	0.355454774	0.287990767	Disintegration	3.1	0.1004	—	—1.168	0	0	0.2880	0.1382
21	0.146330588	0.668451531	Nominal disinteg.	55.3	0.2688	0.1392	—2.848	0.002	—	0.6685	0.1972
22	0.123048894	0.188874396	Nominal disinteg.	101.00	0.0707	0.0629	—1.326	0.006	—	0.1889	0.1202
23	0.093379200	0.033925931	Disintegration	75.4	0.0619	—	—1.206	0.013	0.189	0.3393	0.0669
24	0.391574265	0.032756103	Nominal disinteg.	288.3	0.0335	0.0130	—0.207	0.039	—	0.0328	0.6646
25	0.263022343	0.247991904	Disintegration	13.9	0.0256	—	—1.192	0.003	0.524	0.2480	0.0251
26	0.097623704	0.760908567	Nominal disinteg.	1.9	0.2475	—	—3.231	0.002	—	0.7509	0.0502
27	0.165730470	0.139119704	$\frac{\Delta E}{E}>0.05$	—	—	—	—	—	—	—	—
28	0.112470028	0.564850334	Disintegration	10.2	0.0737	—	—2.444	0	0.612	0.5649	0.0037
29	0.163015720	0.465528719	Disintegration	2.0	0.1133	—	—2.062	0	0.136	0.4655	0.0094
30	0.453271966	0.242887395	Close approach	—	—	—	—	—	—	—	—
31	0.109806163	0.283848681	Disintegration	42.9	0.0725	—	—1.549	0.005	0.207	0.2838	0.0187
32	0.071236454	0.567962233	Nominal disinteg.	7.0	0.1904	—	—2.463	0.012	—	0.4680	0.0304
33	0.048520920	0.290818296	Disintegration	108.5	0.0364	—	—1.602	0.004	0	0.2908	0.4725
34	0.224688133	0.572102054	Disintegration	14.0	0.1701	0.1414	—2.447	0.003	0	0.4721	0.0424
35	0.480736434	0.134853383	Disintegration	6.0	0.0228	—	—3.110	0.001	—0.028	0.1349	0.0927
36	0.101054627	0.091224803	Close approach	—	—	—	—	—	—	—	—
37	0.203387744	0.658870561	Disintegration	1.9	0.1073	—	—2.808	0	0.783	0.6589	0.0366
38	0.236529953	0.551711569	$\frac{\Delta E}{E}>0.05$	—	—	—	—	—	—	—	—
39	0.470810730	0.135874846	Disintegration	124.6	0.0347	0.0160	—3.224	0.001	0.001	0.1359	0.0530
40	0.446076458	0.206502066	Close approach	—	—	—	—	—	—	—	—
41	0.302550914	0.196574135	Disintegration	258.5	0.0263	—	—0.929	0.045	0	0.1966	0.9398
42	0.455307350	0.188735628	Disintegration	63.5	0.0409	—	—0.579	0.018	0.243	0.1887	0.0827
43	0.199431868	0.713511463	Close approach	—	—	—	—	—	—	—	—
44	0.210788097	0.581520233	Disintegration	1.9	0.0968	—	—2.488	0	0.201	0.5815	0
45	0.308297011	0.301011953	$\frac{\Delta E}{E}>0.05$	—	—	—	—	—	—	—	—
46	0.221349313	0.127070459	Close approach	—	—	—	—	—	—	—	—
47	0.330832116	0.495365951	Nominal disinteg.	24.5	0.1396	—	—2.095	0.001	—	0.4954	0.0174
48	0.046961055	0.558717858	Disintegration	41.0	0.1644	—	—2.432	0.001	0	0.5587	0.0097
49	0.057727505	0.167417674	$\frac{\Delta E}{E}>0.05$	—	—	—	—	—	—	—	—
50	0.100298172	0.425877548	Close approach	—	—	—	—	—	—	—	—
51	0.000427526	0.486750493	Nominal disinteg.	45.8	0.0714	—	—2.183	0.009	—	0.4868	0.4145
52	0.055152381	0.426512804	$\frac{\Delta E}{E}>0.05$	—	—	—	—	—	—	—	—

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1968SvA....11.1006A

TABLE 1 (cont.)

N	ξ	η		T	p	p_{min}	$E \cdot 10^{-7}$	$\frac{\Delta E}{E}$	$\frac{\Delta' E}{E}$	$M \cdot 10^{-4}$	$\frac{\Delta M}{M} \cdot 10^{-4}$
1	2	3	4	5	6	7	8	9	10	11	12
53	0.004454218	0.145784306	Disintegration	97.6	0.0665	—	—1.366	0.045	0.016	0.1458	0.1160
54	0.007748064	0.565092305	Disintegration	50.9	0.0507	—	—2.452	0.009	1.289	0.5651	0.3316
55	0.473503305	0.116893831	Disintegration	6.6	0.0360	—	—2.393	0.008	0.018	0.1169	0.0064
56	0.386569322	0.339951672	$\frac{\Delta E}{E}>0.05$	—	—	—	—	—	—	—	—
57	0.101269911	0.306704186	$\frac{\Delta E}{E}>0.05$	—	—	—	—	—	—	—	—
58	0.435627483	0.049049590	$\frac{\Delta E}{E}>0.05$	—	—	—	—	—	—	—	—
59	0.176293211	0.167790923	$\frac{\Delta E}{E}>0.05$	—	—	—	—	—	—	—	—
60	0.305717285	0.373832423	Disintegration	103.2	0.0115	—	—1.592	0.009	6.837	0.3738	0.3491
61	0.437455895	0.150592949	Disintegration	50.7	0.0429	—	—0.429	0.005	2.226	0.1506	0.0591
62	0.068169446	0.038072853	Close approach	—	—	—	—	—	—	—	—
63	0.159779582	0.581268387	Disintegration	7.4	0.1471	—	—2.498	0	0.132	0.5813	0.0141
64	0.359693815	0.224207350	Nominal disinteg.	—	—	—	—0.893	0.001	—	0.2242	0.6231
65	0.306397814	0.264435224	$\frac{\Delta E}{E}>0.05$	—	—	—	—	—	—	—	—
66	0.152497417	0.286843464	Disintegration	62.6	0.0624	0.0368	—1.505	0.002	0.007	0.2868	0.0345
67	0.136803084	0.167749452	Disintegration	139.8	0.0661	0.0195	—1.265	0.011	—0.533	0.1677	0.4580
68	0.258888896	0.567517590	Disintegration	32.7	0.1539	—	—2.421	0.012	0.001	0.5675	0.0025
69	0.378927315	0.238608305	Nominal disinteg.	—	—	—	—0.916	0.014	—	0.2386	0.7573
70	0.386784473	0.009493945	$\frac{\Delta E}{E}>0.05$	—	—	—	—	—	—	—	—
71	0.379442579	0.344937472	Disintegration	15.7	0.0842	0.0755	—1.393	0	0.295	0.3449	0.0348
72	0.150913059	0.733827112	Disintegration	1.9	0.2168	—	—3.120	0.001	0	0.7338	0.0545
73	0.035567328	0.125720780	Disintegration	22.3	0.0484	—	—1.333	0.002	0.692	0.1257	0.1289
74	0.017131688	0.597193411	Close approach	—	—	—	—	—	—	—	—
75	0.171277915	0.163977452	$\frac{\Delta E}{E}>0.05$	—	—	—	—	—	—	—	—
76	0.280145301	0.040746517	Disintegration	42.1	0.0278	—	—0.665	0.011	0.598	0.0407	0.1916
77	0.111245697	0.032869382	Disintegration	114.8	0.0842	0.0779	—1.175	0.032	0	0.0329	1.0243
78	0.129733170	0.710744272	Nominal disinteg.	—	—	—	—3.023	0.001	—	0.7107	0.0366
79	0.177281222	0.520422521	$\frac{\Delta E}{E}>0.05$	—	—	—	—	—	—	—	—
80	0.110426012	0.387555532	Nominal disinteg.	59.9	0.0902	—	—1.836	0.004	—	0.3876	0.0464
81	0.242888665	0.198203787	Disintegration	39.8	0.0436	—	—1.091	0.002	0.715	0.1982	0.0202
82	0.142397675	0.385411857	Disintegration	107.4	0.0555	—	—1.805	0.009	0.166	0.3854	0.3941
83	0.118589788	0.456797180	$\frac{\Delta E}{E}>0.05$	—	—	—	—	—	—	—	—
84	0.184943235	0.016807856	$\frac{\Delta E}{E}>0.05$	—	—	—	—	—	—	—	—
85	0.013188964	0.429651778	Nominal disinteg.	41.6	0.1616	—	—1.999	0.004	—	0.4297	0.1857
86	0.085778906	0.072485138	$\frac{\Delta E}{E}>0.05$	—	—	—	—	—	—	—	—
87	0.450134134	0.230877165	Close approach	—	—	—	—	—	—	—	—
88	0.432697866	0.7311230217	$\frac{\Delta E}{E}>0.05$	—	—	—	—	—	—	—	—
89	0.048364955	0.311528956	Close approach	—	—	—	—	—	—	—	—
90	0.237107399	0.295829168	Close approach	—	—	—	—	—	—	—	—
91	0.199219026	0.217809880	$\frac{\Delta E}{E}>0.05$	—	—	—	—	—	—	—	—
92	0.199254563	0.392218395	Close approach	—	—	—	—	—	—	—	—
93	0.196018586	0.471972279	Nominal disinteg.	182.2	0.2093	0.1119	—2.066	0.005	—	0.4720	0.1316
94	0.144736081	0.248008566	Disintegration	2.2	0.0134	—	—1.417	0.003	1.218	0.2480	0.0013
95	0.139575010	0.447612270	Close approach	—	—	—	—	—	—	—	—
96	0.468459592	0.180165843	Disintegration	5.0	0.0696	—	—0.528	0.016	0	0.1802	0.1165

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1968SvA....11.1006A

TABLE 1 (cont.)

N	ξ	η		T	p	p_{min}	$E \cdot 10^{-7}$	$\frac{\Delta E}{E}$	$\frac{\Delta' E}{E}$	$M \cdot 10^{-4}$	$\frac{\Delta M}{M} \cdot 10^{-4}$
1	2	3	4	5	6	7	8	9	10	11	12
97	0.347295515	0.249211481	Disintegration	3.3	0.0445	—	−1.019	0.001	0.899	0.2492	0.0578
98	0.174891833	0.530514918	Disintegration	2.0	0.1142	—	−2.297	0	0.443	0.5305	0.0217
99	0.255720500	0.025414176	$\frac{\Delta E}{E} > 0.05$	—	—	—	—	—	—	—	—
100	0.007783970	0.488557558	$\frac{\Delta E}{E} > 0.05$	—	—	—	—	—	—	—	—

a series of approaches of the components, of which the last is a close triple approach, the system disintegrates, one of the components recedes, and the other two form a double system.

As Table 1 indicates, the disintegration time fluctuates over a wide range, depending on the initial conditions. Figure 3 shows a normalized distribution curve for the disintegration time, as prepared from the data of Table 1.

From the 59 values for the disintegration time we find that its mean value is

$$\bar{T} = 61.6, \tag{12}$$

and the rms deviation from the mean is

$$\sigma = 74.4. \tag{13}$$

Thus we may write approximately (since T does not follow a normal distribution)

$$\bar{T} = 62 \pm 10. \tag{14}$$

All these quantities are expressed in units of τ, the mean time for crossing the system.

Consequently, if the components of the system have the same mass as the sun (k = 1) and if the distances between the components at initial time are approximately 0.03 pc ($l = 3$), with small velocities at that time, then according to Eqs. (1), (7), and (14) the mean disintegration time will be $(1.6 \pm 0.3) \cdot 10^6$ yr, with an rms deviation of $1.9 \cdot 10^6$ yr.

If the triple system consists of hot giants with masses equal to $10m_\odot$, then for the same distances between components at initial time the mean disintegration time of the system would be $(5.1 \pm 0.8) \cdot 10^5$ yr, with an rms deviation of $7 \cdot 10^5$ yr.

These values for the time intervals may be regarded as characteristic values for the disintegration time of triple systems in stellar cosmogony.

In the case of triple galaxies, if the mass of each galaxy is $2 \cdot 10^{43}$ g and the distances separating them are 30 kpc, then the mean disintegration time will be $(1.6 \pm 0.3) \cdot 10^{10}$ yr, with an rms deviation of $1.9 \cdot 10^{10}$ yr.

The characteristic value of the disintegration time of triple systems having negative energy in the cosmogony of galaxies is 16 billion years. Thus most triple systems of galaxies with negative energy will not be capable of breaking up within a cosmogonically acceptable period.

Figure 4 illustrates the position of the body A_3 for the 62 initial times for which the computation of trajectories has been completed. Each point is labeled with the corresponding value obtained for

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1968SvA....11.1006A

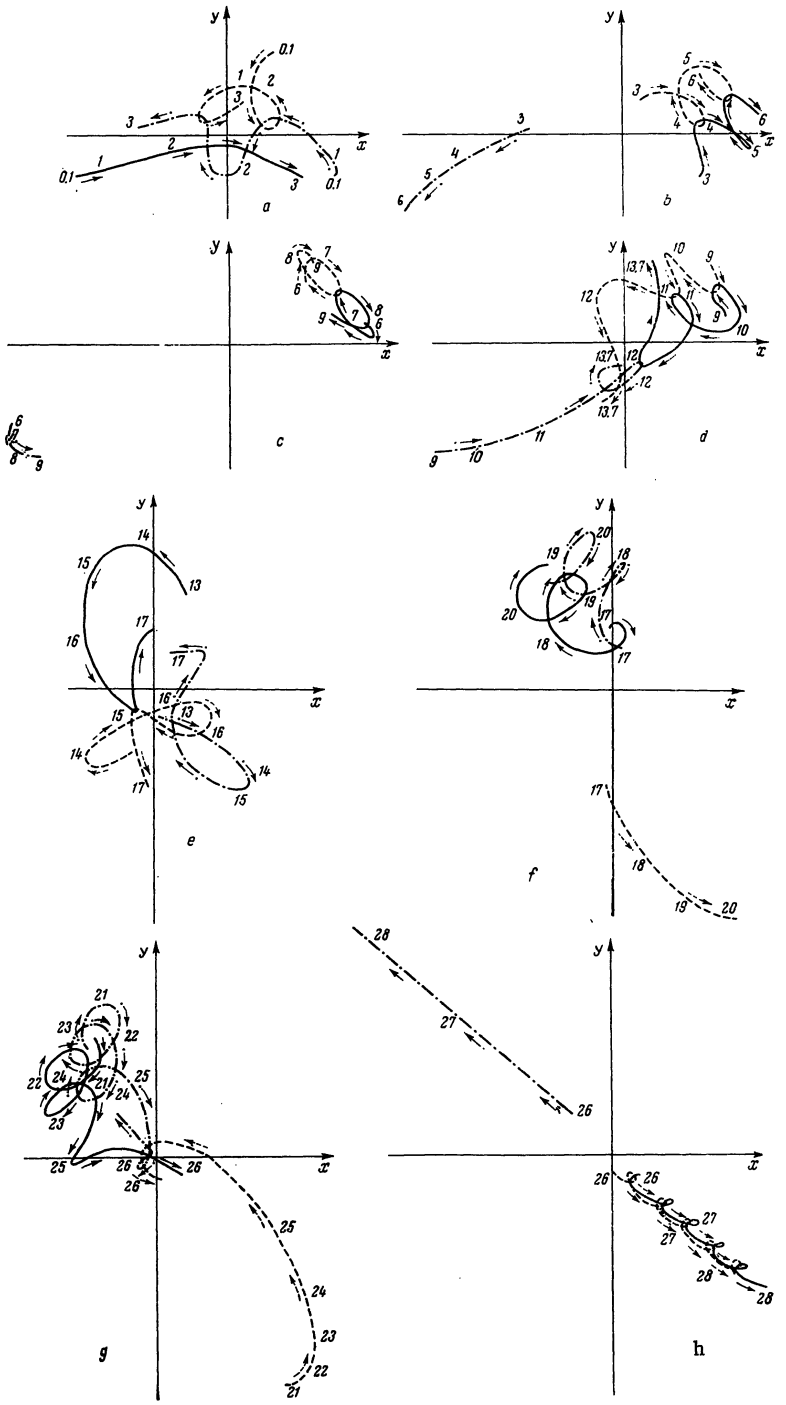


Fig. 2

the disintegration time, expressed in units of τ. It is of particular interest to examine whether the disintegration time is a function of the coordinates ξ and η. The data of Fig. 4 reveal sharp changes in the value of T for comparatively small changes in ξ and η, and indicate the absence of any graphic general law for the variation of T.

These results could have been foreseen in advance. Of course the 62 points plotted in Fig. 4 are too few for us to attempt to discover any regular dependences of T on ξ and η with their aid. It might happen that regions exist (of very small area), whether simply or multiply connected, within which T is a continuous function of the position of

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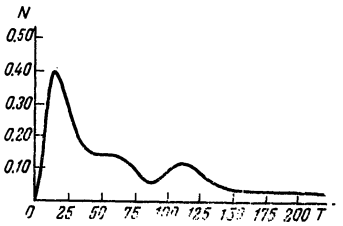


Fig. 3

the point A_3. At the boundaries of such a region T would suffer a discontinuity. Suppose that for some ξ and η a system were to disintegrate after a certain triple approach and a disintegration time of T. Let ξ and η be subjected to infinitesimally small increments $\Delta\xi$ and $\Delta\eta$. If in the new triple system disintegration were to occur for the same approach of the components as in the previous system, then the corresponding ΔT would also be infinitesimally small. But if disintegration were to occur in the new triple system as a result of another approach of the components, then ΔT would be finite, so that the function T would lose its discontinuity at the point (ξ, η). Is there only a limited number of regions within which the function T is continuous? We would conjecture that the number of regions is unlimited.

Another important question concerns the existence of initial states not leading to disintegration — those for which $T = \infty$. We may at once point out that initial states in which the positions of the components and their velocities are symmetric relative to the center of inertia of the system cannot result in disintegration, for in this event the symmetry of the positions should be preserved throughout the course of the motion, whereas disintegration is an asymmetric phenomenon: One component recedes, while the other two remain bound to each other.

Once initial positions with $T = \infty$ exist the question arises whether it is admissible, as is done in this investigation, to find a mean value for T. The answer should be affirmative. In the first place, the mean value of T is determined for different ξ and η, but for the same initial velocities as prescribed by Eqs. (3). One will see that under the conditions (3) the velocities cannot be symmetric relative to the center of inertia. But even in the event that all possible initial values for the velocities of the components were considered, it would still be admissible to establish $\overline{T}$, since one could claim that the set of initial conditions symmetric relative to the center of inertia of the system is a set of measure zero in comparison with the set of all possible initial conditions.

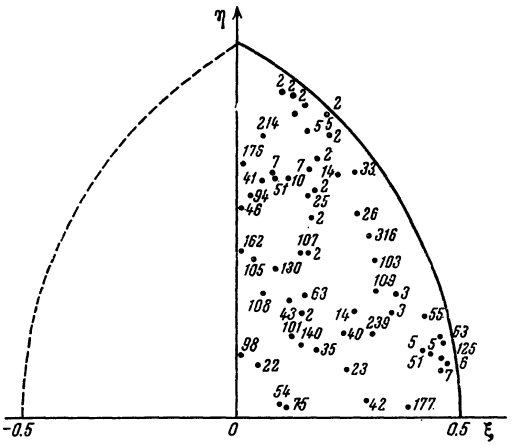


Fig. 4

It is known that a triple system can be stable and will not disintegrate if two neighboring components describe mutually elliptic motions while a third remote component describes an elliptic motion about the center of inertia of the first two components. Clearly the initial conditions (3) adopted in the present investigation exclude the possibility of such a motion. It is therefore admissible to find $\overline{T}$ under the conditions (3). In the general case, one would be able to find $\overline{T}$ by excluding from consideration in some appropriate manner the collection of initial conditions corresponding to stable systems.

The investigation described in this paper will be extended, according to current plans, to include the cases of: 1) other values for the angular momentum of the system; 2) different masses for the components of the system; 3) the three-dimensional problem.

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